

Single-port BRAM 50k bit (Uses Block Memory)

```
//Verilog Code

`timescale 1ns / 1ps

module BRAM_SinglePort(

    input wire clk,

    input wire en,

    input wire we,

    input wire [10:0] addr,

    input wire [31:0] din,

    output reg [31:0] dout);

    (* ram_style = "block" *) reg [31:0] mem [0:1562];

    always @(posedge clk) begin

        if (en) begin

            if (we)

                mem[addr] <= din;

            dout <= mem[addr];

        end

    end

endmodule
```

1. Slice Logic

Site Type	Used	Fixed	Available	Util%
Slice LUTs*	0	0	53200	0.00
LUT as Logic	0	0	53200	0.00
LUT as Memory	0	0	17400	0.00
Slice Registers	0	0	106400	0.00
Register as Flip Flop	0	0	106400	0.00
Register as Latch	0	0	106400	0.00
F7 Muxes	0	0	26600	0.00
F8 Muxes	0	0	13300	0.00

2. Memory

Site Type	Used	Fixed	Available	Util%
Block RAM Tile	2	0	140	1.43
RAMB36/FIFO*	2	0	140	1.43
RAMB36E1 only	2			
RAMB18	0	0	280	0.00

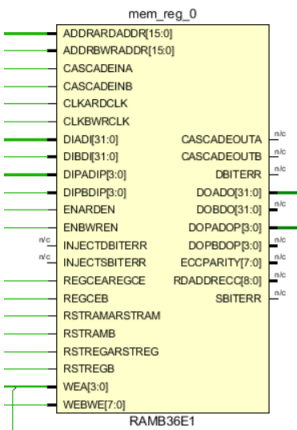


Figure 1: Block memory 1

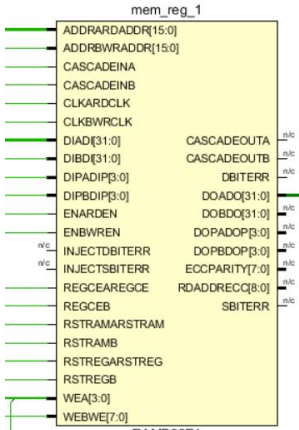


Figure 2: Block memory 2

RTL analysis Schematic:

